

# Aayush

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LeetCode: 400+

## Profile Summary

B.Tech CSE student (CGPA 9.43) with hands-on ML experience spanning from-scratch implementations, fine-tuned transformers, and production-grade RAG systems. Strong mathematical foundations in probability, statistics, and linear algebra. 400+ LeetCode problems solved. Seeking to deepen applied ML expertise through Amazon ML Summer School 2026.

## Education

### Bachelor of Technology in Computer Science and Engineering

2023 – 2027

I.K. Gujral Punjab Technical University CGPA: **9.43** *Relevant Coursework: Probability*

*Statistics, Linear Algebra, Optimization, DBMS*

## Projects

### ML Algorithms from Scratch using NumPy \*

- Implemented Linear Regression, Logistic Regression, and a feedforward Neural Network using only NumPy; manually derived gradient descent and backpropagation including MSE/Cross-Entropy losses and L2 regularization
- Benchmarked against Scikit-learn on Iris and Boston Housing datasets, achieving within 2% accuracy; validated generalization via k-fold cross-validation and bias-variance tradeoff analysis
- Tech Stack:** Python, NumPy, Matplotlib, Gradient Descent, Backpropagation, Cross-Validation

### Enterprise RAG AI Assistant \*

- Built a RAG-based assistant over 1,000+ PDF pages using semantic search; optimized FAISS vector pipeline across 50,000+ embeddings, cutting retrieval latency from 4.8s to under 1.5s
- Implemented conversational memory, improving multi-turn query accuracy by 35% over baseline LLM responses; supported 10+ file formats with citation-aware responses
- Tech Stack:** Python, LangChain, FAISS, ChromaDB, Gemini/OpenAI API, Streamlit, Vector Databases

### Sentiment Analysis with Fine-Tuned DistilBERT \*

- Fine-tuned DistilBERT for binary sentiment classification on 50K+ reviews; achieved 91% F1-score (5-fold CV), a 15% improvement over TF-IDF + Logistic Regression baseline
- Built full PyTorch training pipeline with tokenization, confusion matrix evaluation; published model on Hugging Face Hub
- Tech Stack:** Python, PyTorch, Hugging Face Transformers, Scikit-learn, Pandas, NLP

## Skills

- ML & Deep Learning:** Supervised/Unsupervised Learning, Gradient Descent, Backpropagation, Bias-Variance Tradeoff, Cross-Validation, Random Forest, XGBoost, CNNs, Transformers, Feature Engineering
- Mathematics for AI:** Probability, Statistics, Hypothesis Testing, Linear Algebra, Optimization, Calculus
- GenAI & NLP:** PyTorch, TensorFlow, Hugging Face Transformers, RAG, FAISS, Embeddings, Prompt Engineering, LangChain
- Data Science:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, EDA, Data Preprocessing
- Programming & Tools:** Python, Java, C++, SQL, MySQL, Git, Jupyter, Linux, Kaggle

## Achievements & Leadership

- Kaggle Competition — Titanic: Machine Learning from Disaster** — Achieved top 8% ranking (global leaderboard) with an ensemble of Random Forest + XGBoost; applied advanced feature engineering including family size, title extraction, and cabin deck encoding
- Solved **400+ DSA problems** on LeetCode with an active contribution streak; strong focus on graphs, DP, and binary search
- District Topper** in a National-Level Examination
- Ranked 1st** in Robotics and Artificial Intelligence branch
- Campus Executive, **E-Cell IIT Bombay** — led cross-functional teams for technical events and workshops
- Member, **Training Advisory Council (TAC)** — collaborated with faculty on curriculum development initiatives

## Certifications

- Building AI-Powered Search with MongoDB Vector Search** – MongoDB \*
- LLMs on Google Cloud: Retrieval-Augmented Generation on Vertex AI** – Infosys \*
- Introduction to Natural Language Processing** – Infosys \*